Device-Related Thrombus after LAA Occlusion in Case 3: Image Assessment and Haemodynamic Risk

Hugo Jal

Modelling of Organs and Systems, Universitat Pompeu Fabra

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Abstract

Is there a device-related thrombus (DRT)? No. In the CT, contrast-filled blood reaches the atrial face of the occluder and no thrombus can be seen on it. **Is the patient at risk of developing one?** Yes. A flow simulation of the left atrium (LA) with physiological flow and a mitral valve moving along its axis shows that the device sits in the quietest part of the atrium: the blood around it moves slowly, and the wall next to it has the lowest and most oscillating wall shear stress, which are the conditions linked to thrombus formation.

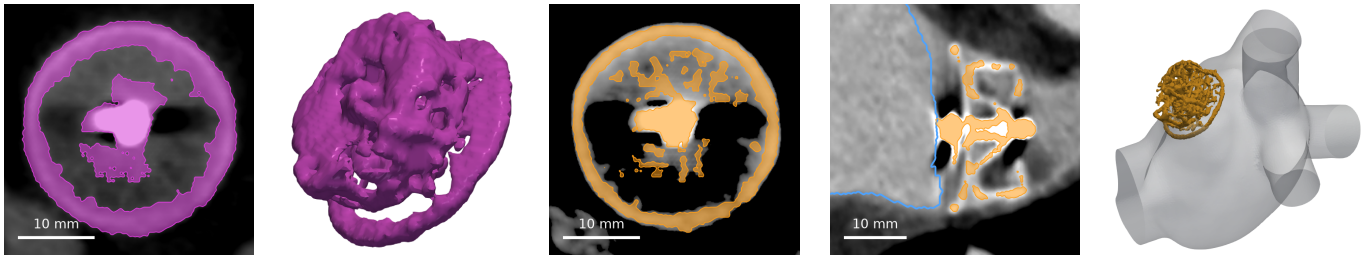


Figure 1: Segmentation. (a) U-Net device segmentation (magenta) on the plane of the device disc and (b) its 3D model. (c) Gamma-transformed CT on the same plane with the wire-frame segmentation (orange): rim, struts and hub. (d) CT along the device axis (LA on the left) with the LA (blue) and wire frame (orange): bright contrast blood reaches the atrial face of the device. (e) LA model (transparent) and device wire frame.

1 Introduction

Left atrial appendage occlusion (LAAO) closes the appendage with a device to prevent strokes in patients with atrial fibrillation. A known complication is a thrombus growing on the atrial face of the device. This device-related thrombus (DRT) is not frequent, but patients who develop it have a clearly higher risk of stroke [1]. In CT, a DRT appears as a dark (hypoattenuating) layer on the atrial side of the device [2]. Patient-specific flow simulations have linked DRT to slow blood and to low, oscillating wall shear stress around the device [3, 4]. In this report I answer two questions for Case 3: is there a DRT, and is the patient at risk of one?

2 Methods

2.1 Segmentation and mesh

The contrast CT (diastolic phase) was segmented in 3D Slicer (Fig. 1). The LA was obtained by manual thresholding of the contrast-filled blood, and then refined with manual painting and erasing, using the threshold range as a mask. The four pulmonary veins (PVs) were kept up to their first branch, the LA was closed at the mitral valve (MV), and the device was left out of the blood volume. The device was segmented manually, and I tried two ways to extract only its wire frame. First, a pretrained promptable 3D U-Net (nnInteractive [6]), prompted with points on bright wires and on the blood around them. Alone it also filled the blood between the wires, so its output was intersected with the local wire contrast of the CT and corrected in 3D Slicer: it finds the rim and the hubs, but the struts merge into thick bands (Fig. 1a,b). Second, a gamma transform of the CT ($\gamma = 0.3$) and a filter for thin bright lines gave thinner struts and a closed rim (Fig. 1c). The surface was smoothed and repaired in MeshLab and Meshmixer, and the PVs were cut flat and extended with short cylinders. The MV was cut on the annulus plane found in the CT and extended 10 mm along the valve axis, which was made the z axis. The volume was meshed with tetrahedra in Gmsh (308 000 elements).

2.2 Flow simulation

The flow was solved in Ansys Fluent as transient and laminar, with blood as a Newtonian fluid ($\rho = 1060\,\mathrm{kg/m^3}$, $\mu = 3.5\,\mathrm{mPa\,s}$). The PVs received the course pressure curve. The MV received the course velocity curve and its longitudinal motion (up to 10 mm in systole), applied along the valve axis with a dynamic mesh (smoothing and remeshing). On the wide annulus the unchanged curve would give 130 mL per beat, so I scaled it to a normal cardiac output of 5.0 L/min (73 mL per beat) [7], keeping its shape. With the time step of the boundary-condition files (0.01 s) I simulated four heartbeats and analysed only the last, free of start-up effects. The residuals converged at every step, and the last two heartbeats gave the same flow volumes and almost the same wall maps.

2.3 Analysis

Results were exported at every time step and visualised in ParaView. From the wall shear stress $\boldsymbol{\tau}_w$ over one heartbeat of length T I computed

$$\mathrm{TAWSS} = \tfrac{1}{T}\int_0^T |\boldsymbol{\tau}_w|\,dt,$$

$$\mathrm{OSI} = \tfrac{1}{2}\Big(1 - \frac{\left|\int_0^T \boldsymbol{\tau}_w\,dt\right|}{\int_0^T |\boldsymbol{\tau}_w|\,dt}\Big), \quad \mathrm{ECAP} = \frac{\mathrm{OSI}}{\mathrm{TAWSS}}.$$

TAWSS is the average shear on the wall, OSI says how much the shear changes direction (0: never, 0.5: half of

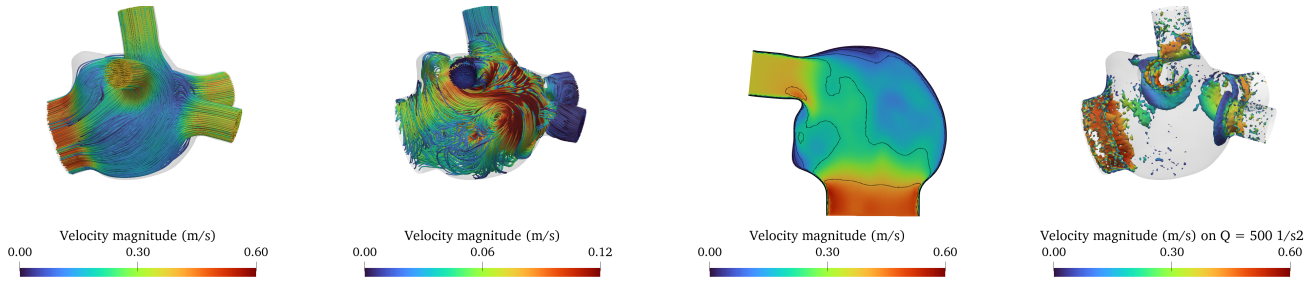


Figure 2: Flow in the last heartbeat. Streamlines from the pulmonary veins (a) at the peak of mitral flow and (b) in late diastole. (c) Velocity at the peak on a plane through the valve axis (valve at the bottom) and the device, which closes the bulge on the left wall. (d) Vortices at the peak ($Q = 500\,\mathrm{s}^{-2}$ [5]).

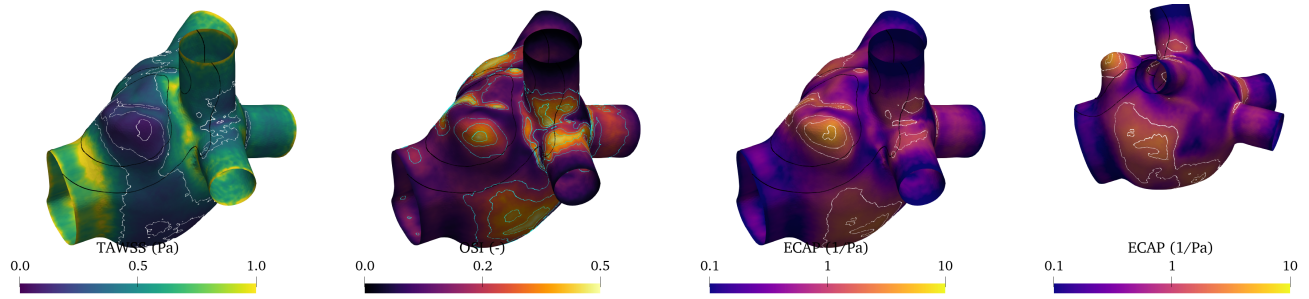


Figure 3: Wall shear metrics over the last heartbeat, seen from the device side (a–c) and from the front (d): (a) TAWSS, (b) OSI, (c, d) ECAP (log scale), each with isolines. Black lines: 10 and 20 mm from the device.

the time), and ECAP is high where the shear is both low and oscillating, which is the condition linked to thrombus formation [4]. Because the mesh changes as the valve moves, the wall shear of each time step was interpolated onto one fixed copy of the wall before averaging. To compare the device region with the rest of the atrium, I grouped the wall by distance to the device. For the flow I looked at streamlines, velocity on a plane through the device and vortex structures, shown with a Q-criterion of $500\,\mathrm{s}^{-2}$ as in left-atrium 4D flow MRI [5].

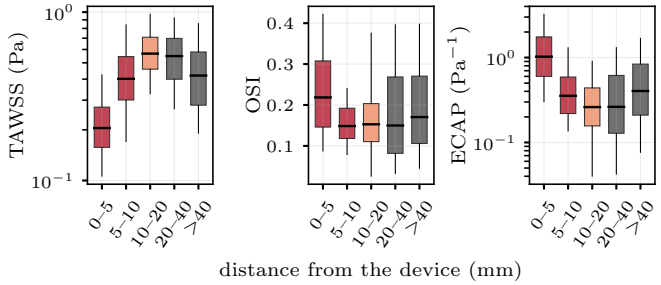


Figure 4: TAWSS, OSI and ECAP grouped by distance from the device (box: interquartile range, line: median, whiskers: 5th–95th percentile). Red: near the device.

3 Results

3.1 Is there a thrombus?

No. In the CT slices the atrial face of the device is in direct contact with bright, contrast-filled blood (Fig. 1d). There is no dark layer covering the device, so no thrombus is visible.

3.2 Flow near the device

The flow is physiological: 73 mL per beat, an even inflow from the four veins and velocities below 0.6 m/s. During rapid filling, the streams from the four PVs meet in the middle of the atrium and form a jet towards the MV, with vortex rings at the vein inlets (Fig. 2a,d). The device lies at the edge of the cavity, away from this path, and the blood at the entrance of the occluded appendage is the slowest of the section even at the peak of filling (Fig. 2c). In late diastole the flow slows down to one large vortex in the atrial body, and the streamlines next to the appendage are again among the slowest of the atrium (Fig. 2b), so the blood there is renewed slowly during the whole heartbeat.

3.3 Wall shear metrics

The wall maps (Fig. 3) show one clear region of low TAWSS and high OSI, and therefore high ECAP, centred on the entrance of the appendage where the device sits. The highest ECAP of the whole atrium is found there, inside the 10 mm line. A second, milder high-ECAP region covers part of the lower wall (Fig. 3d). Grouping the wall by distance (Fig. 4) shows the same trend: within 5 mm of the device the shear is the lowest and the most oscillating, and the median ECAP is about three times higher than in the rest of the atrium. Moving away from the device, all three metrics return to the values of the rest of the wall within about 10 mm.

4 Discussion and conclusion

In this atrium, slow blood, low shear and oscillating shear all occur together on and around the atrial face of the device. Patient-specific simulations associate this pattern with a higher DRT risk [3], and in patients who did develop a DRT the thrombi appeared where these maps predicted [4]. The CT shows that no thrombus has formed yet, but this does not mean that the conditions for one are absent.

The model has limitations: the boundary conditions are generic course curves, not measured for this patient, which also set the absolute values, and the device is not in the flow domain as a solid object. For this reason I compared regions within the same atrium rather than using fixed thresholds from the literature.

Conclusion. There is no device-related thrombus in the CT, but **the patient is at risk of developing one**: the device sits in the region with the slowest blood and the lowest, most oscillating wall shear of the whole atrium. Follow-up imaging of the device would be advisable.

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